Profile Azure Benchmark 2024

FIS Profile V7.6.8 / GT.M V7.1-006

RedHat Enterprise Linux 9.4

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Performance Testing Objectives

- Demonstrate Performance and Scalability in the Public Cloud
- Conduct Profile Core Performance Tests on Azure Linux VM Sizes
- Analyze Profile V7.6.8 / GT.M 7.1-006 Performance
  - Deploy and Test High End Premium SSD and Ultra Disk Storage
  - Staged Profile Performance Configurations and Test Cases
    - Buildout of 5M, 10M and 25M Customer/Account Databases
    - Testing Includes Online Simulation and Key Overnight Batch Jobs
    - Encrypted Database
    - Spanned Global Regions
    - Online GT.M Database Backup
    - Increased Online Virtual Users and Batch Thread Analysis
    - Initial AMD Performance Test
    - Initial MTM-SSL Enabled Test
  - Validation of Profile/RHEL 9 Operational Readiness Checklist
  - Document Current Results (Performance Summary and Benchmark Report)

Performance Overall Summary

- Deploying FIS Profile in the public cloud provides robust performance and scalability, making it a viable option for a supported production architecture.
- Achieved 10,000+ Online TPS processing using larger Azure VM Size and Ultra Disk Configuration with average response times of 3 milliseconds.
- Increased Azure VM Type Sizes with additional CPU and memory resources allows for further online throughput and better batch runtime performance.
- Larger Azure VM Types allow higher maximum IOPS and throughput, improving both online and batch performance.
- Upgrading from Standard SSD to Premium SSD (V2) to Ultra Disks allows for increased IOPS and throughput improving both online and batch performance.
- Improved disk storage layout, including multiple striped disks per volume group and filesystem, increases the overall online throughput and batch runtime performance.
- Database Encryption Overhead observed at 2%-6% as Compared to a Standard Non-Encrypted Database.
- MTM-SSL Enabled observed ~5% decrease in performance compared to a Standard MTM.
- Preferred Linux Tuned-ADM setting is set to latency performance.

Azure VM/RHEL 9.4 Configurations

Azure D-series VMs are designed for general-purpose workloads and offer a balanced CPU-to-memory ratio. They are considered versatile and suitable for a wide range of applications.

D16ds_v5

- 8 X 2.8GHz Intel Xeon Platinum 8370C CPUs
- Hyperthreads = 2
- 16 vCPUs
- 64 GB Memory
- RHEL 9.4

D32ds_v5

- 16 X 2.8GHz Intel Xeon Platinum 8370C CPUs
- Hyperthreads = 2
- 32 vCPUs
- 128 GB Memory
- RHEL 9.4

D48ds_v5

- 24 X 2.8GHz Intel Xeon Platinum 8370C CPUs
- Hyperthreads = 2
- 48 vCPUs
- 192 GB Memory
- RHEL 9.4

D96ds_v5

- 48 X 2.8GHz Intel Xeon Platinum 8370C CPUs
- Hyperthreads = 2
- 96 vCPUs
- 384 GB Memory
- RHEL 9.4

Azure Storage/Disk Configuration

Standard SSD Storage

- 500 Max IOPS/100 MBps Throughput

XFS File System Layout

/profile	200 GB
/gbls	500 GB
/journals	500 GB
/backup	2000 GB

Premium SSD V2 Storage

- 3000 Max IOPS/125 MBps Throughput
- 4500 Max IOPS/200 MBps Throughput

XFS File System Layout

/profile_gbls1	200 GB
/profile_gbls2	200 GB
/profile_gbls3	200 GB
/profile_gbls4	200 GB
/profile_gbls5	200 GB
/profile_gbls6	250 GB
/profile_gbls7	250 GB
/profile_gbls8	250 GB
/profile_gbls9	250 GB
/profile_journals1	200 GB
/profile_journals2	200 GB
/profile_journals3	200 GB
/profile_journals4	200 GB
/profile_journals5	200 GB
/profile_journals6	200 GB
/profile_spool	50 GB
/profile_backup	1000 GB

ULTRA Disk Storage

- 3000 Max IOPS/125 MBps Throughput
- 4500 Max IOPS/200 MBps Throughput

XFS File System Layout

/profile_gbls11	500 GB
/profile_gbls12	500 GB
/profile_journals11	250 GB
/profile_journals12	250 GB
/profile_backup11	500 GB

Profile Configuration

- Profile V7.6.8
  - Profile Application: master_v76_6241_42ef7
  - Profile Framework: master_v40_5388_3f66
- GT.M V7.1-006
- Customer and Account Database Sizes

Class	Type	5M Database	10M Database	25M Database
CIF	Personal	5,000,000	10,000,000	25,000,000

Class	Group	Type	5M Database	10M Database	25M Database
Deposit	SAV	300	1,000,000	2,000,000	5,000,000
Deposit	CD	350	500,000	1,000,000	2,500,000
Deposit	DDA	400	2,000,000	4,000,000	10,000,000
Loan	LN	500	500,000	1,000,000	2,500,000
Loan	RC	600	500,000	1,000,000	2,500,000
Loan	MTG	700	500,000	1,000,000	2,500,000
		Total	5,000,000	10,000,000	25,000,000

Online and Batch Workloads

- Online Workload
  - 50% Financial Transactions (DDA and SAV deposit/withdrawals TSSP)
  - 50% Inquiry Messages (Customer and Account Inquiry MSQL)
  - Processed 1,000,000+ online transactions per test

Type	Service Class	Message Percentage
DDA Deposit	TSSP	15%
DDA Withdrawal	TSSP	15%
SAV Deposit	TSSP	10%
SAV Withdrawal	TSSP	10%
Customer Inquiry	MSQL	25%
Account Inquiry	MSQL	25%

- Batch Processing
  - Financial Batch Posting (@QUE066)
  - Interest Posting (@QUE055)
  - Accrual Processing (@QUE039)

Function	Description	Record Count 5M Database	Record Count 10M Database	Record Count 25M Database
@QUE066	Batch Posting	500,000	1,000,000	2,500,000
@QUE055	Interest Posting	150,000	300,000	1,500,000
@QUE039	Accrual Processing	5,000,000	10,000,000	25,000,000

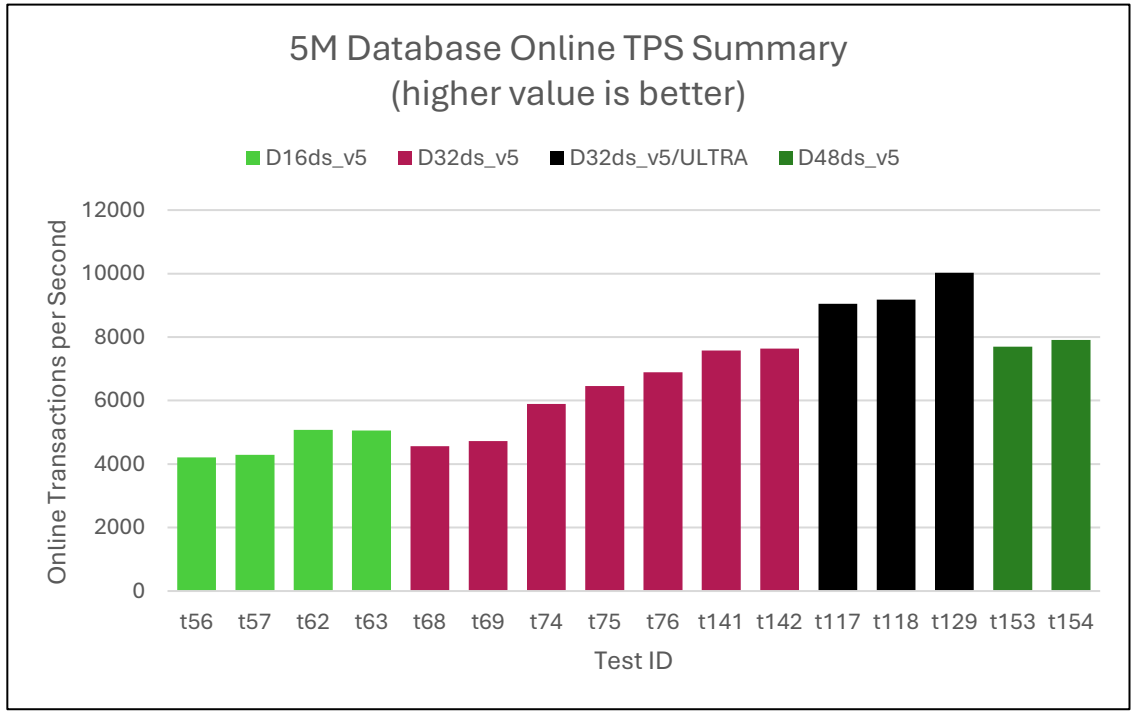
Note – running key critical path batch jobs to evaluate Profile Performance

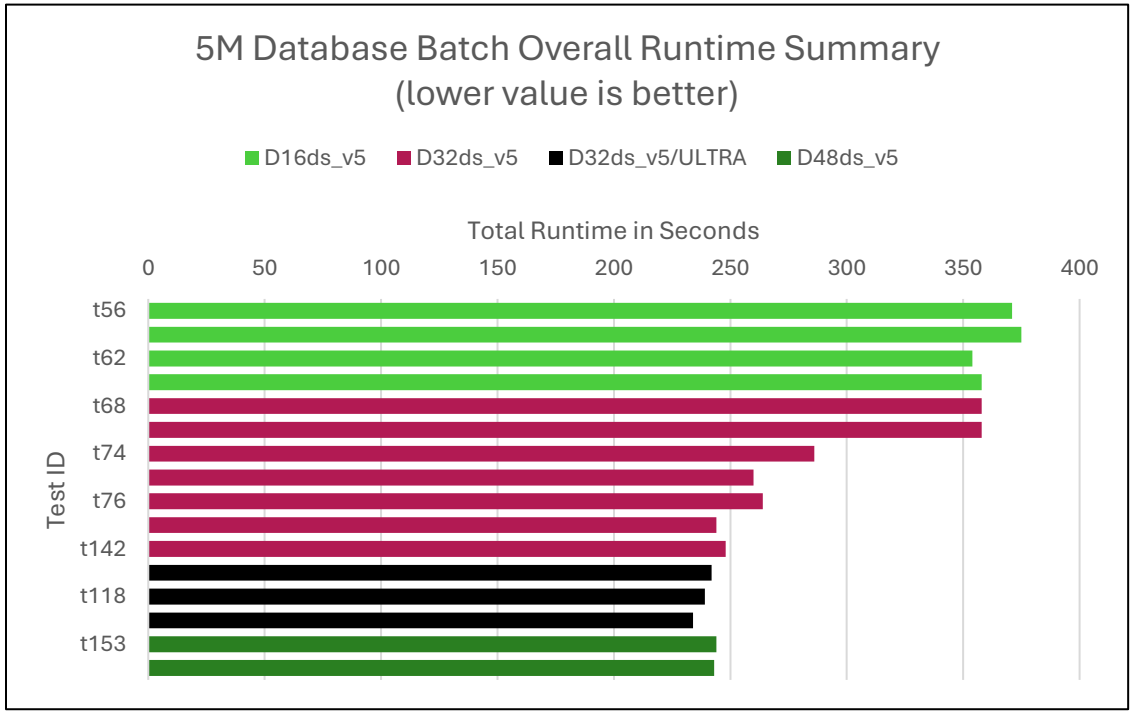
Profile Summary Results

5M Database Size

As test IDs increase , using larger VM sizes and adding processing threads.

- Increased Virtual Users for Online Simulation
- Increased Thread Count for batch processing



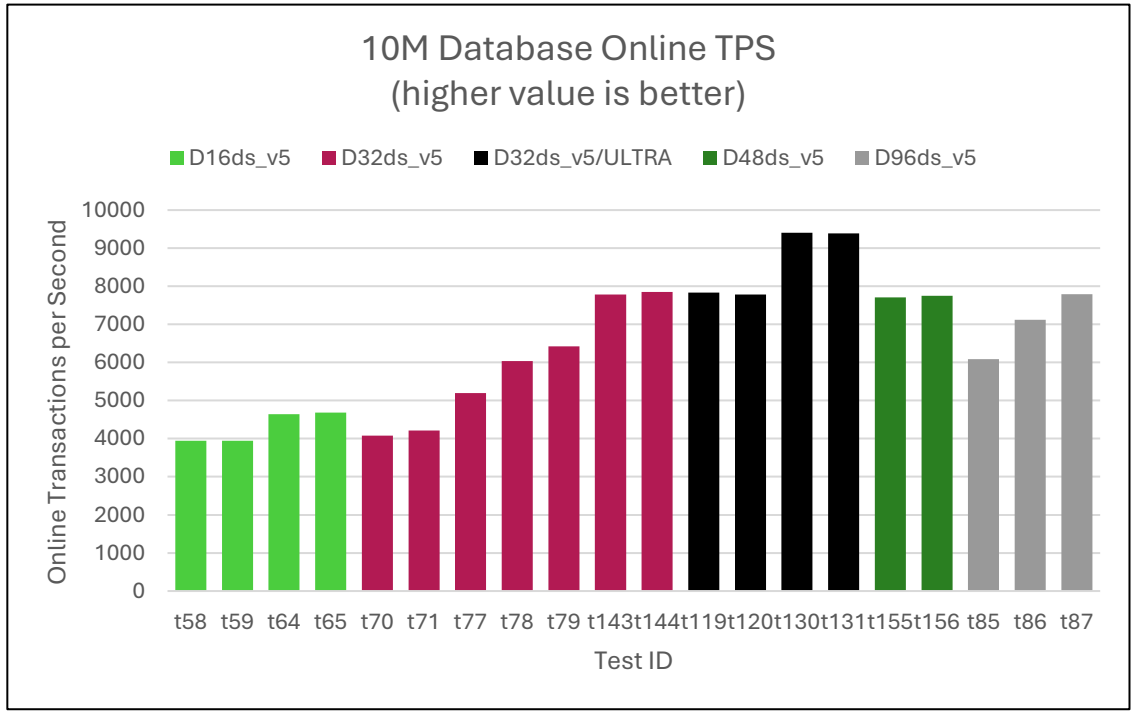


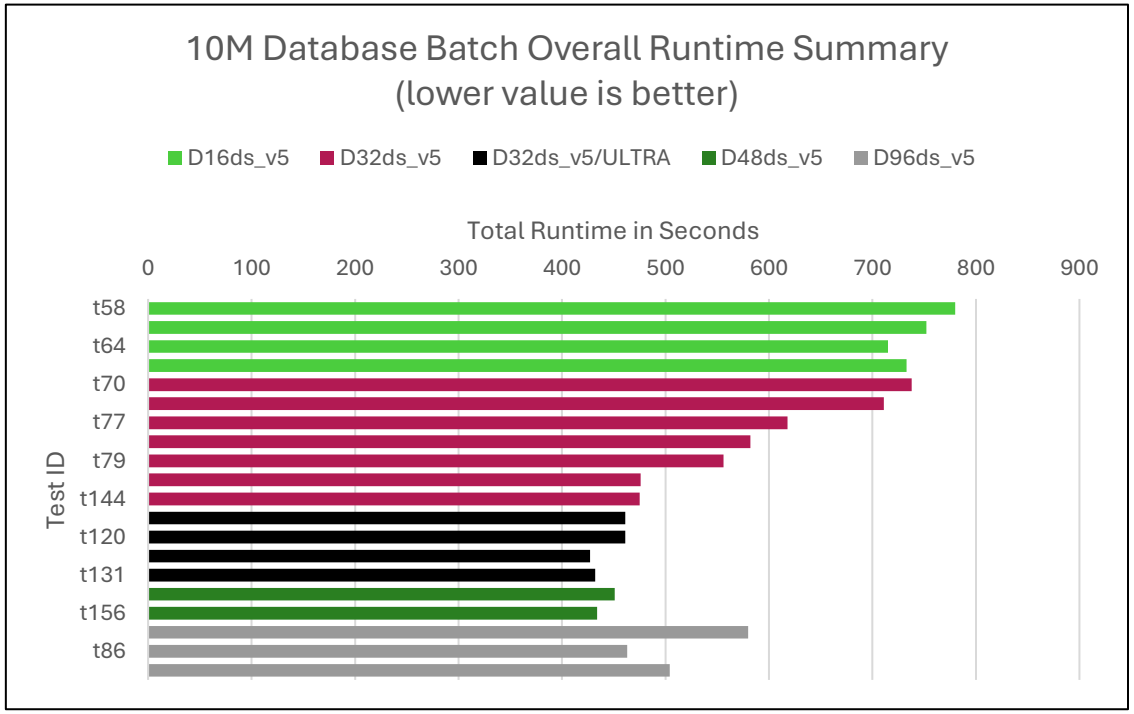
Profile Summary Results

10M Database Size

As test IDs increase , using larger VM sizes and adding processing threads.

- Increased Virtual Users for Online Simulation
- Increased Thread Count for batch processing



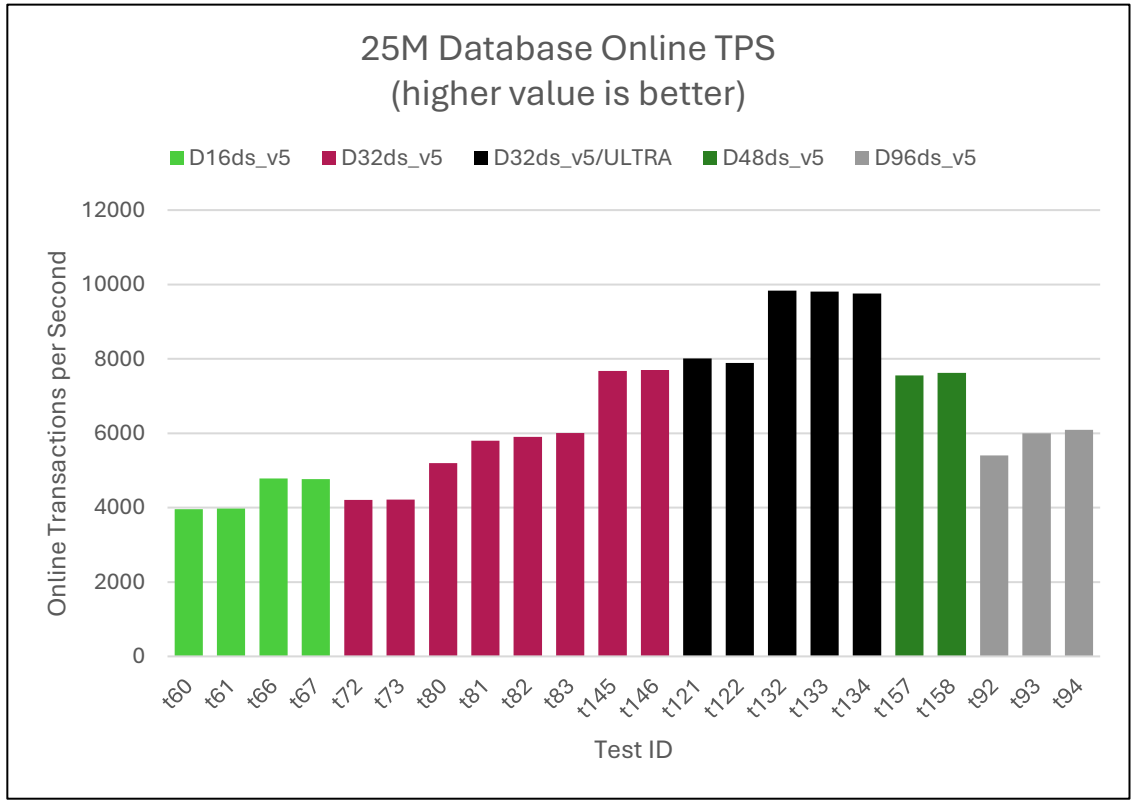


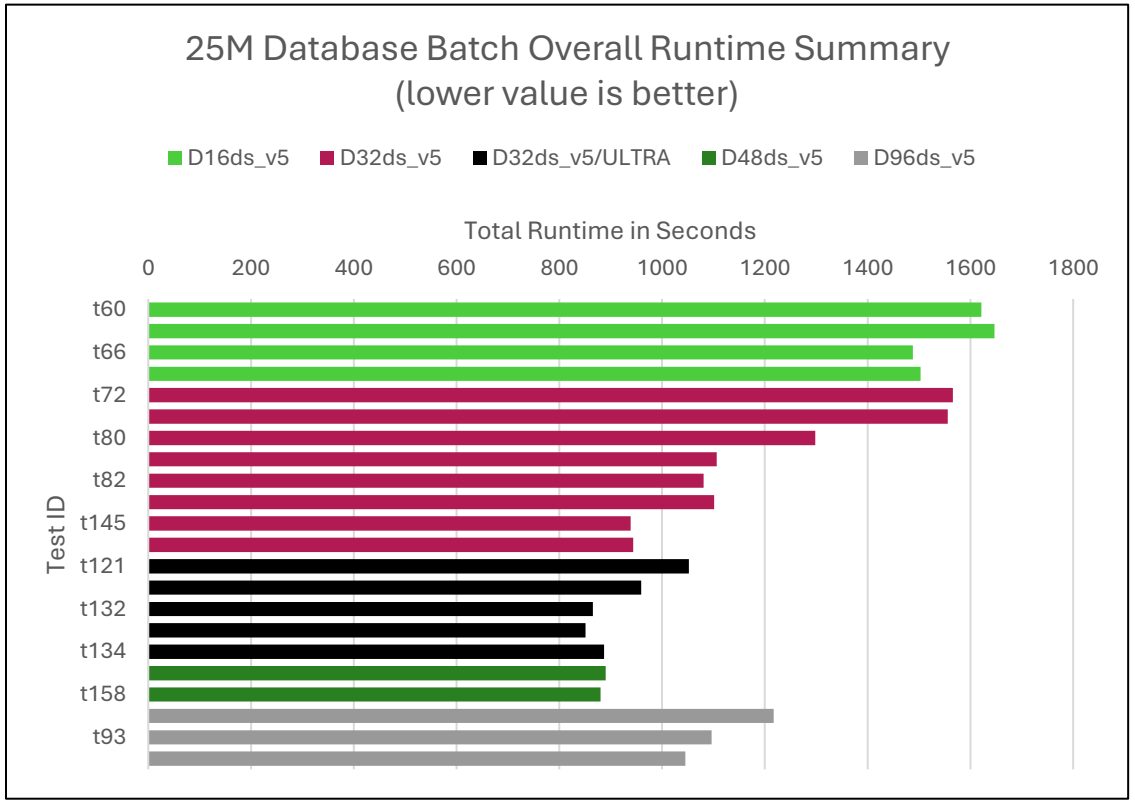
Profile Summary Results

25M Database Size

As test IDs increase , using larger VM sizes and adding processing threads.

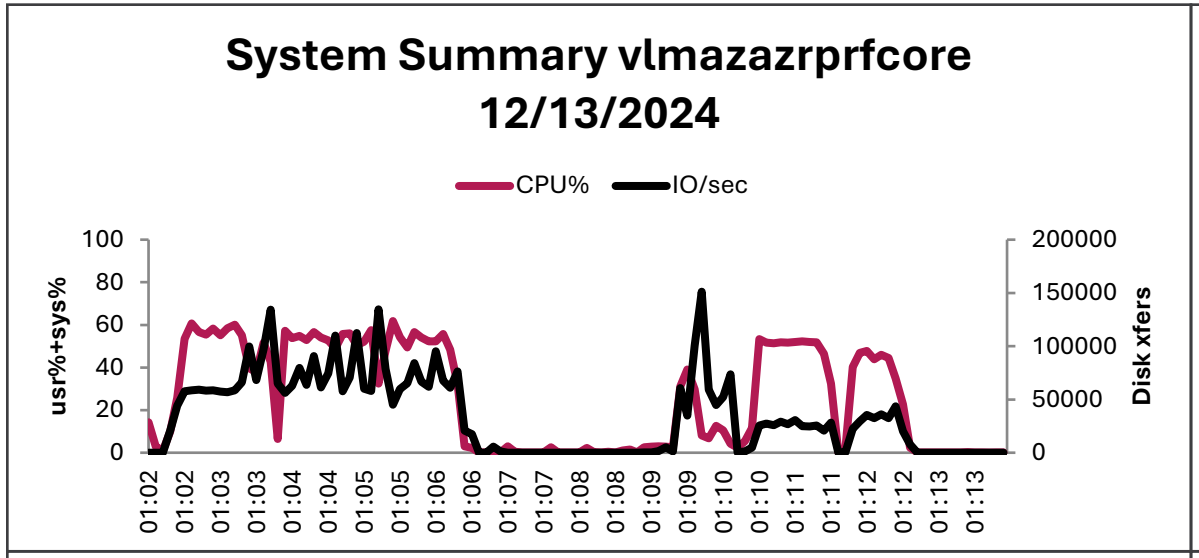
- Increased Virtual Users for Online Simulation
- Increased Thread Count for batch processing

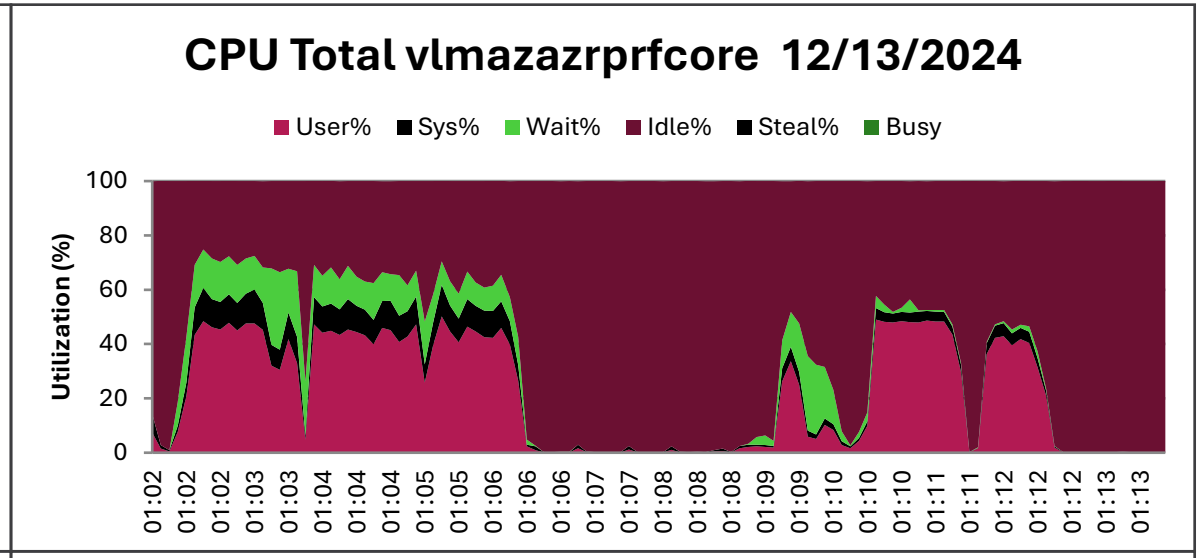


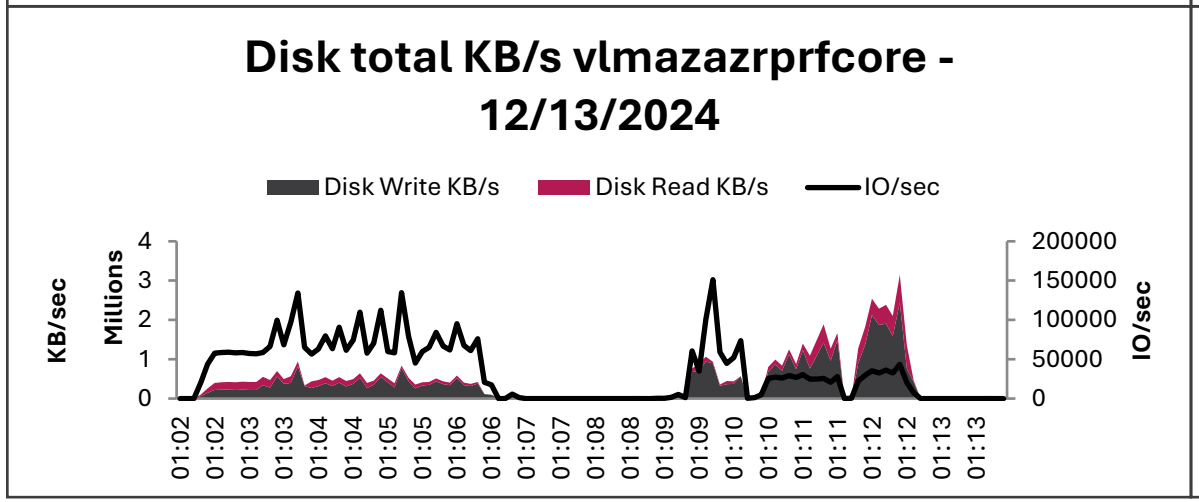


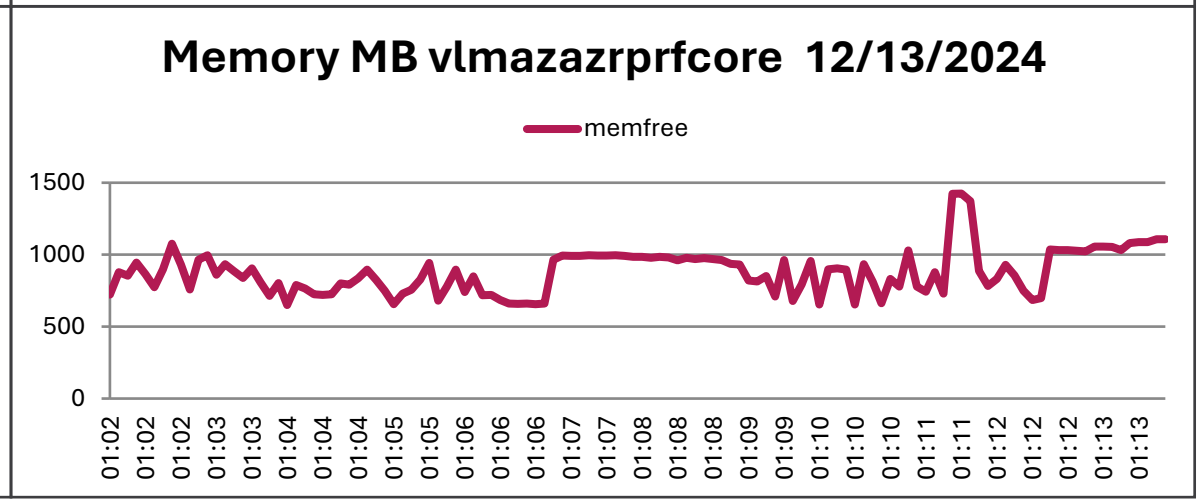
Host Server Utilization Statistics (example)

5M Database Size – t142 D32ds_v5



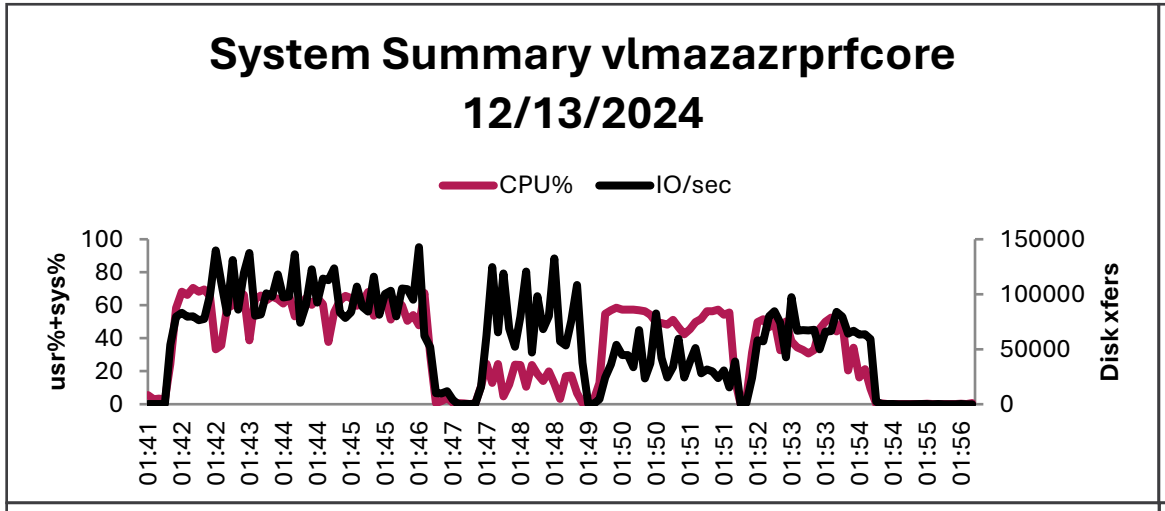


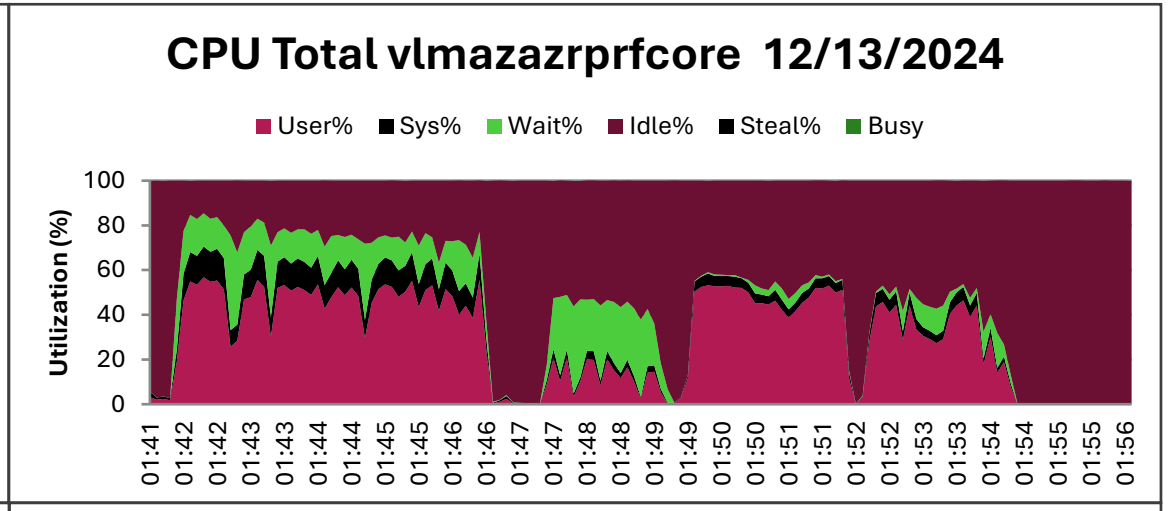


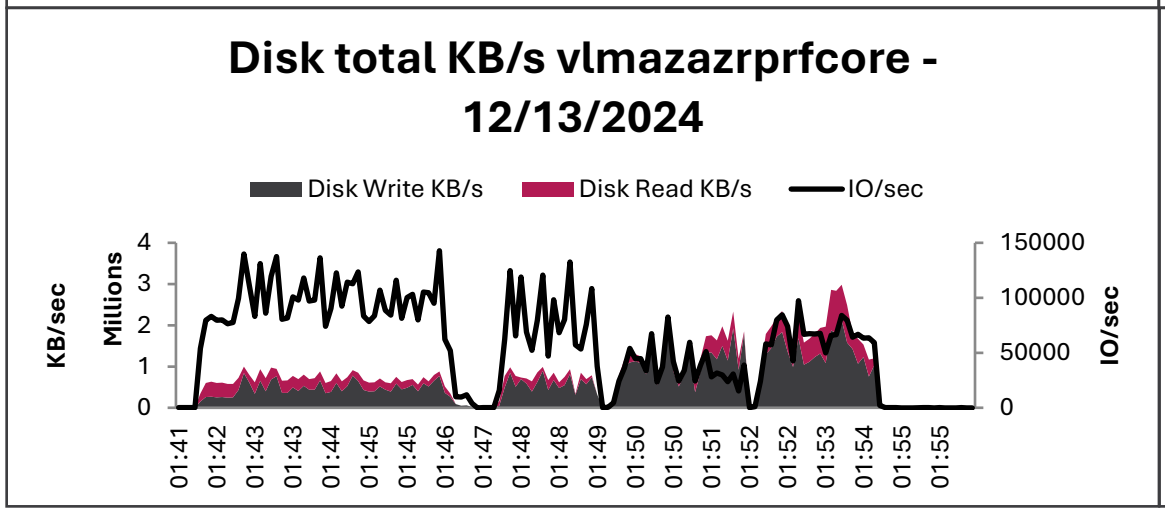


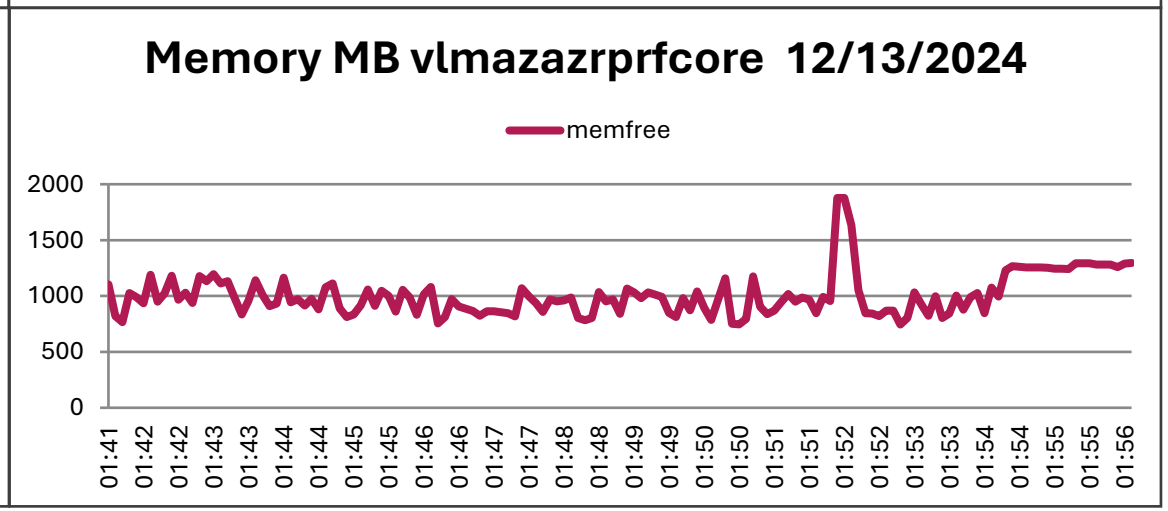
Host Server Utilization Statistics (example)

10M Database Size - t144 D32ds_v5



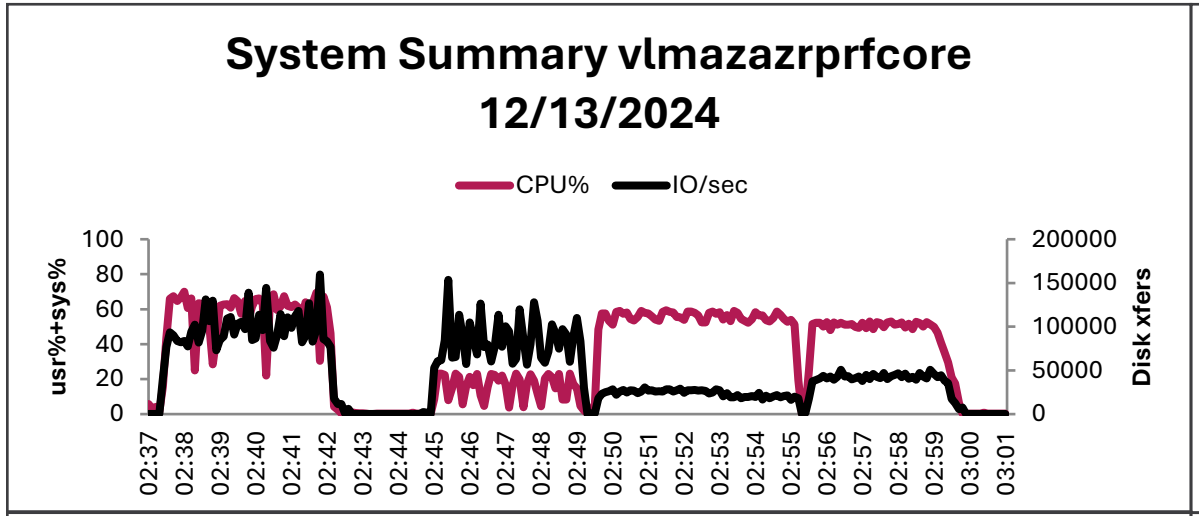


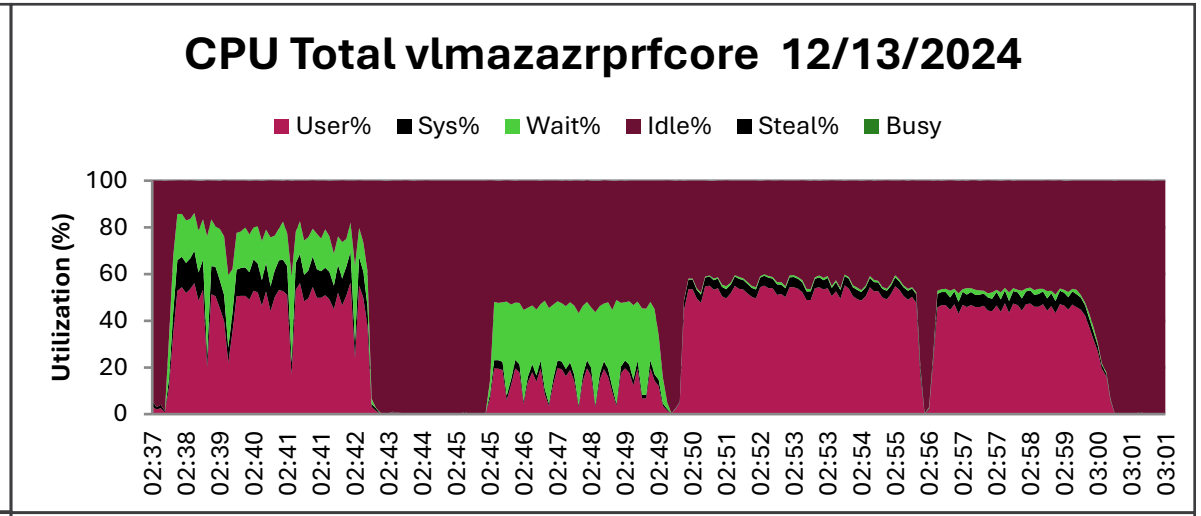


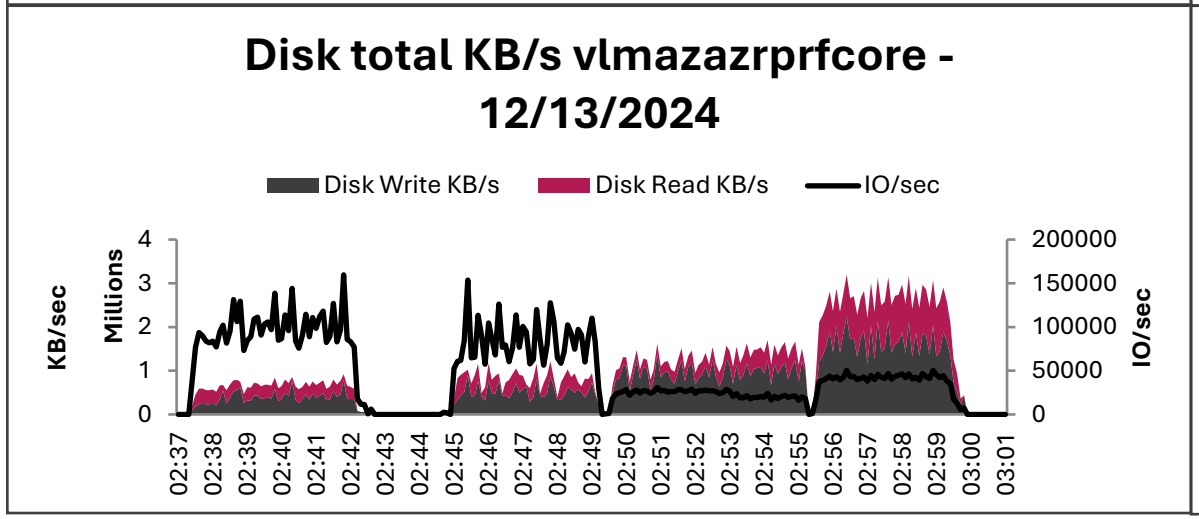


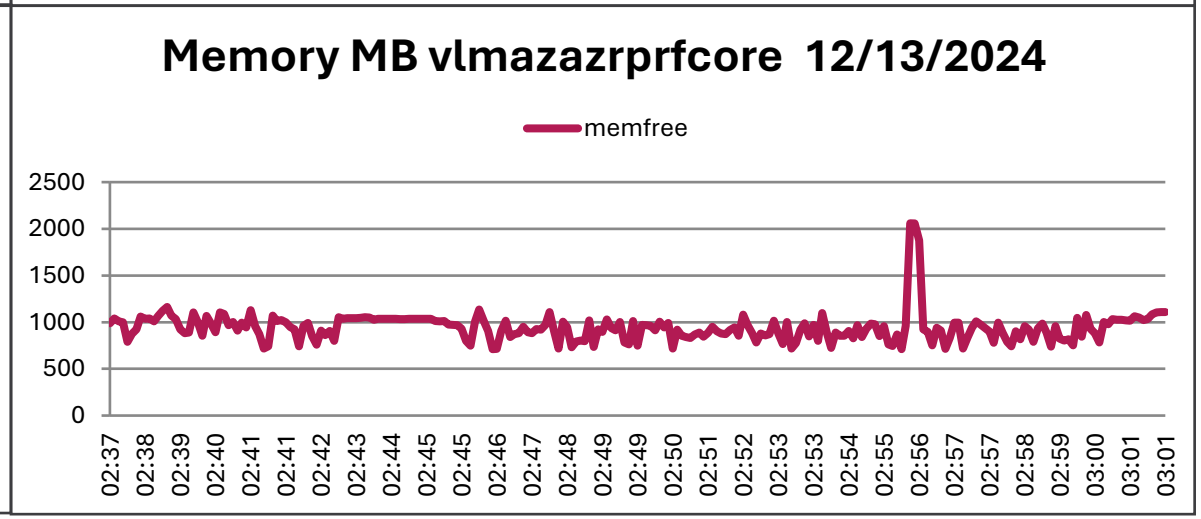
Host Server Utilization Statistics (example)

25M Database Size - t146 D32ds_v5



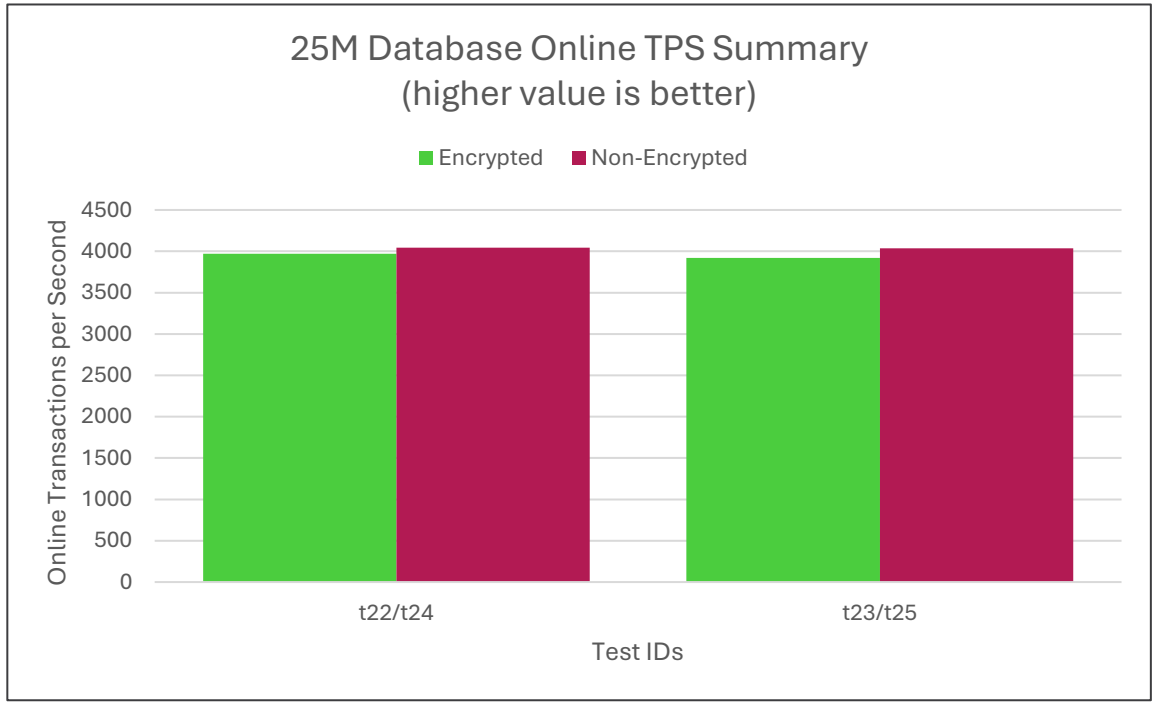


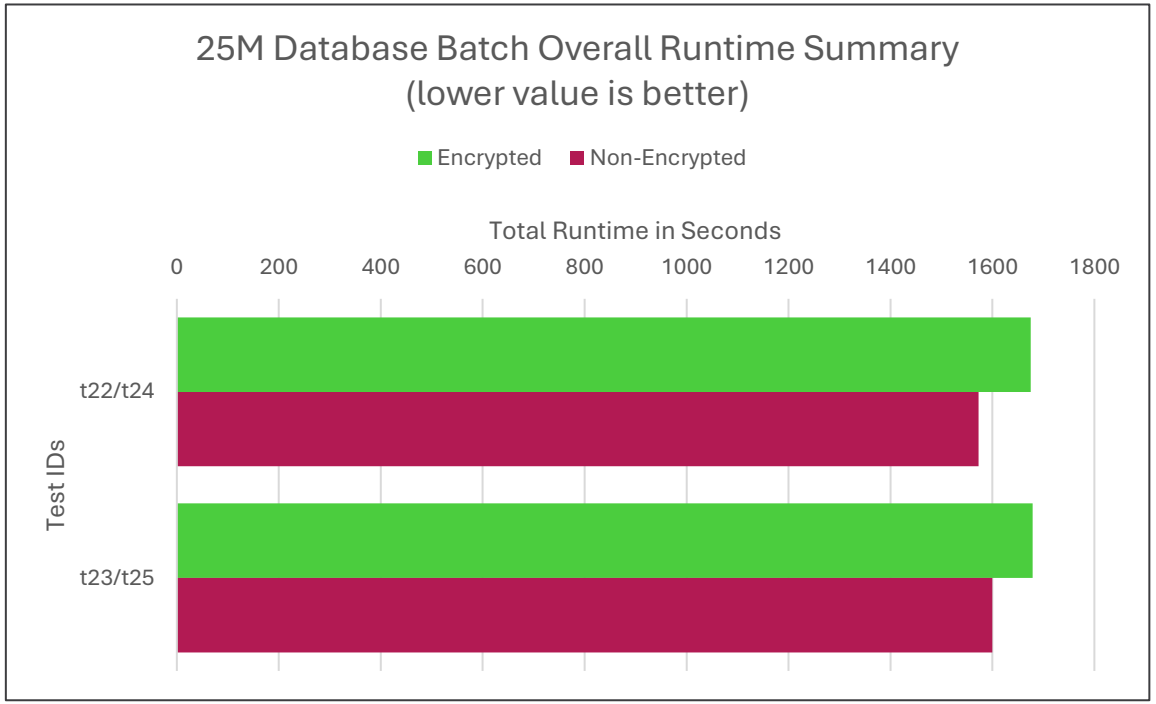




Database Encryption Findings

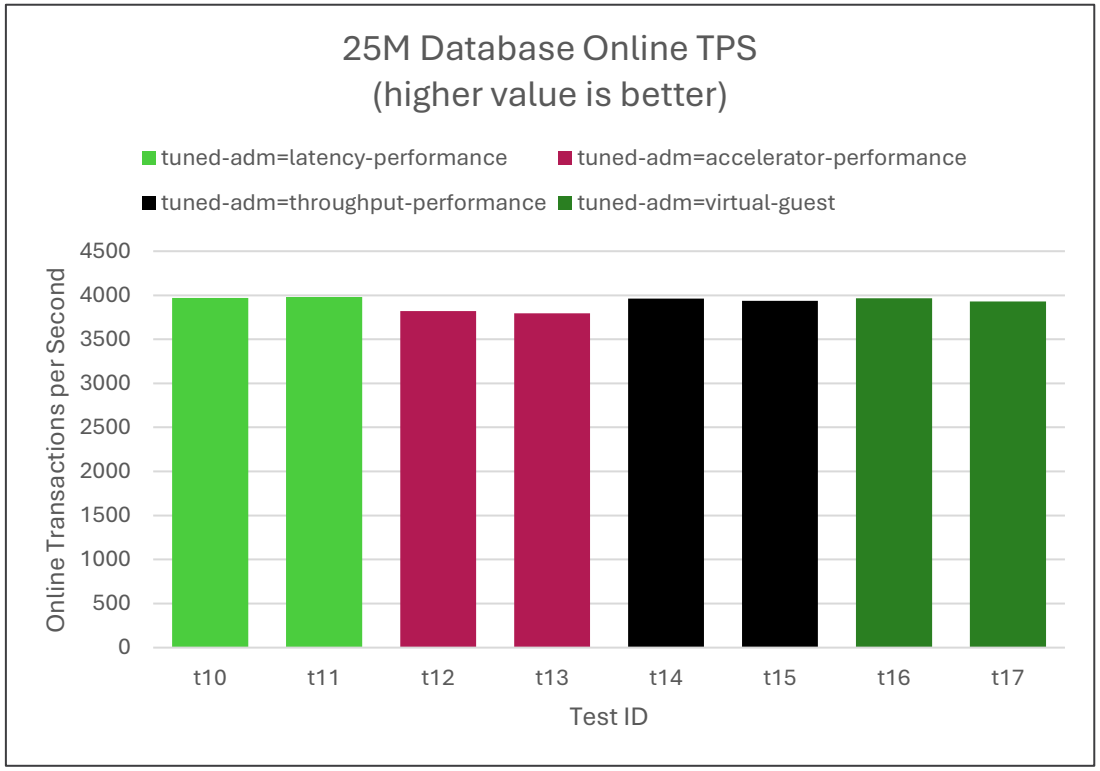
- Encryption Enabled Impacts (AES 256 Plug-in Library Architecture)
  - Online processing shows decrease of ~3% in online throughput performance
  - Batch processing shows decrease of ~6% in overall batch runtime performance

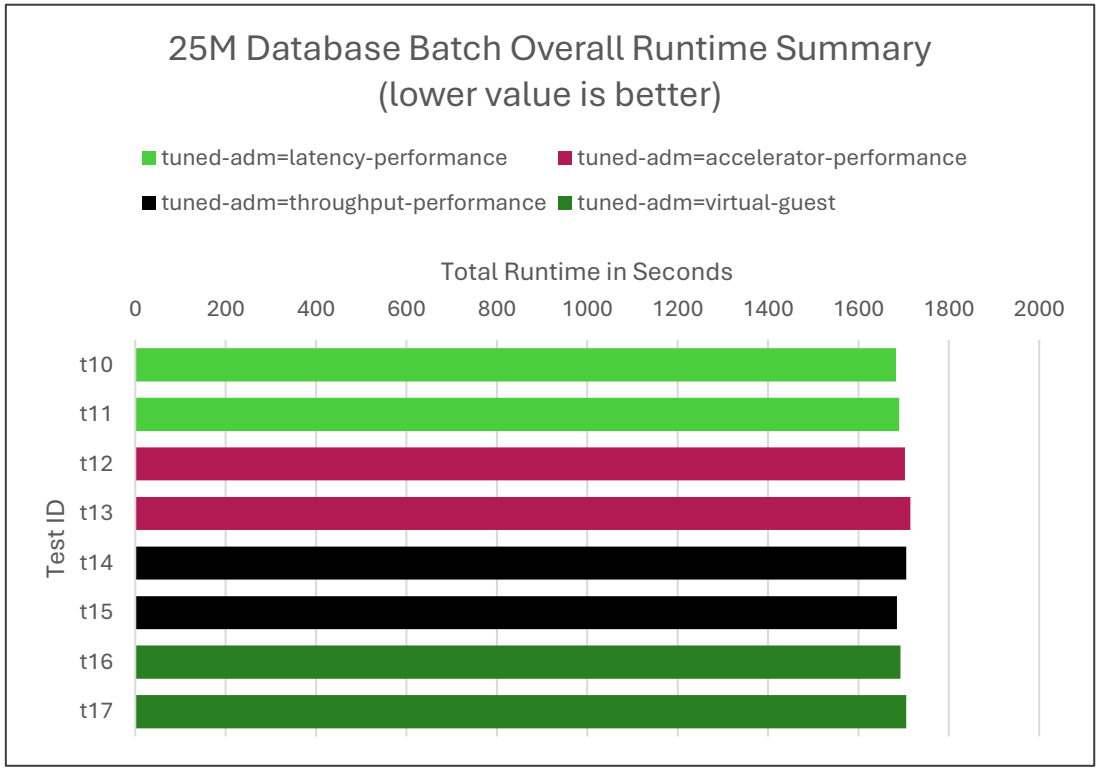




Tuned-ADM Findings

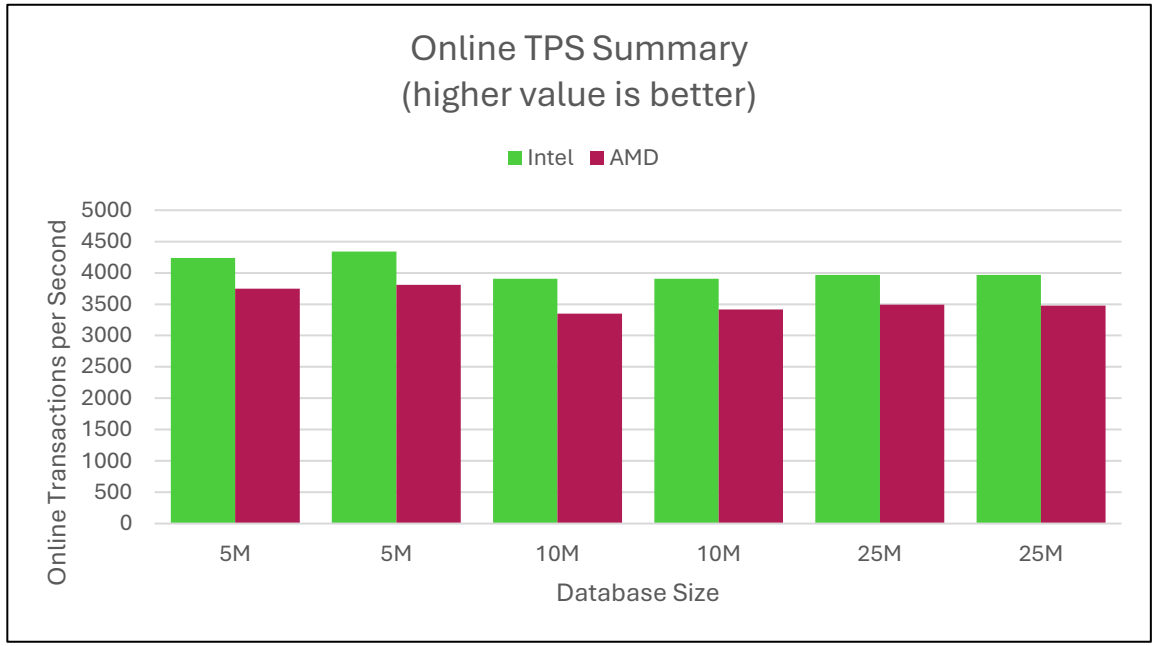
- tuned-adm best practice for Profile
  - current profile=latency-performance

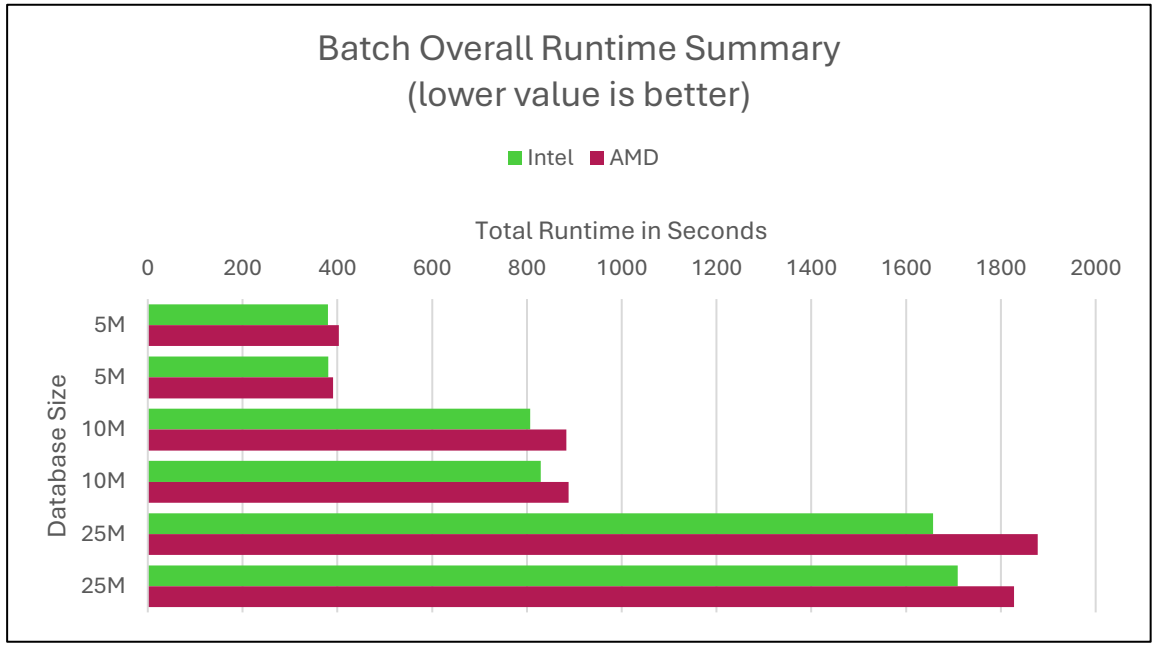




Intel/AMD Findings

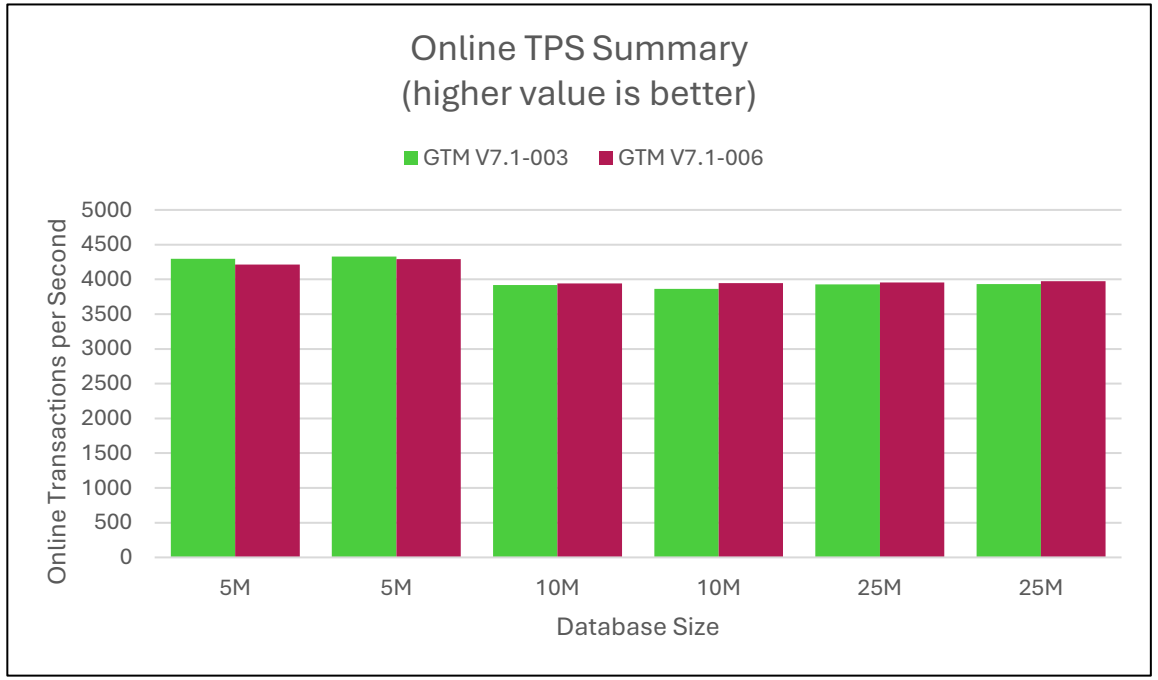
- Intel Xeon Platinum 8370C – 8 x 2.80 GHz CPU (hyperthreads=2)
- AMD EPYC 7763 – 8 x 2.45 GHz CPU (hyperthreads=2)

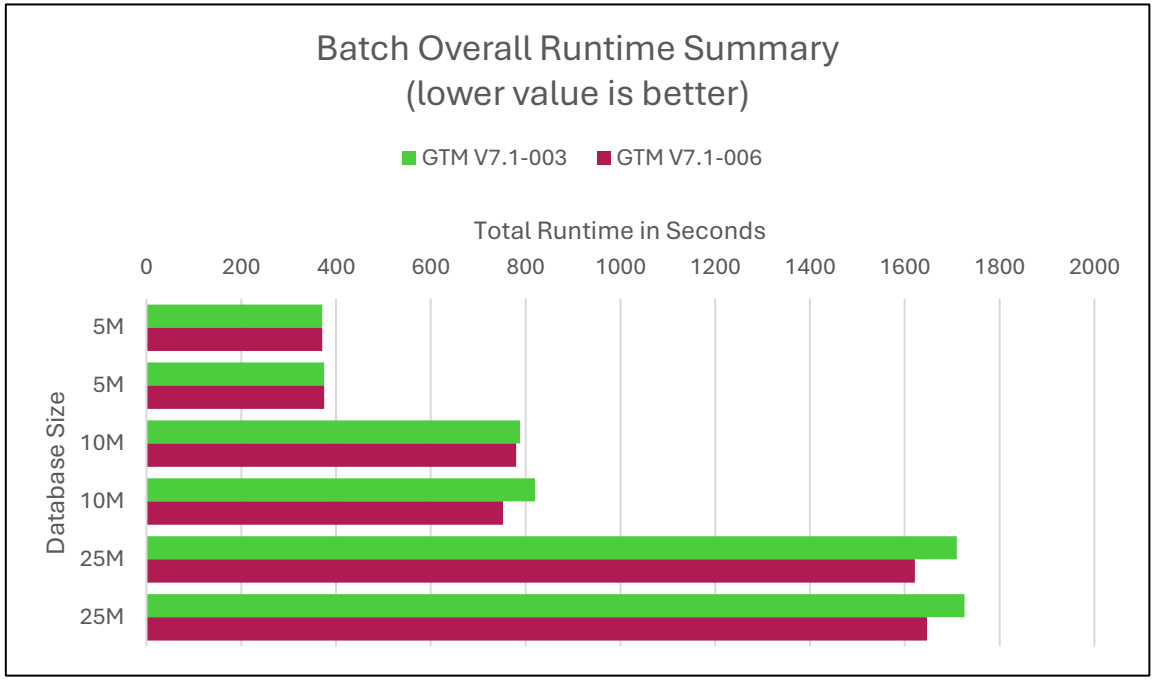




GT.M Build Comparison Findings

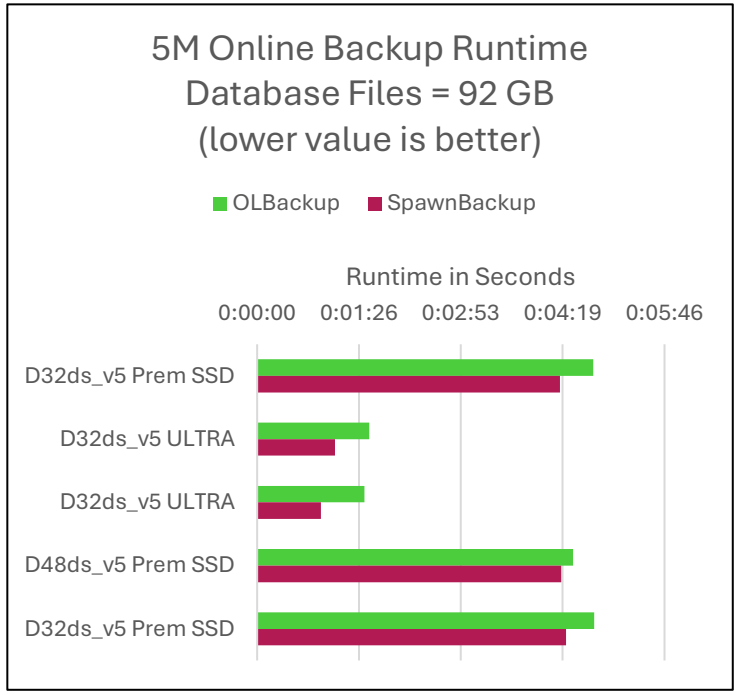
- GT.M V7.1-006 latest general available release version

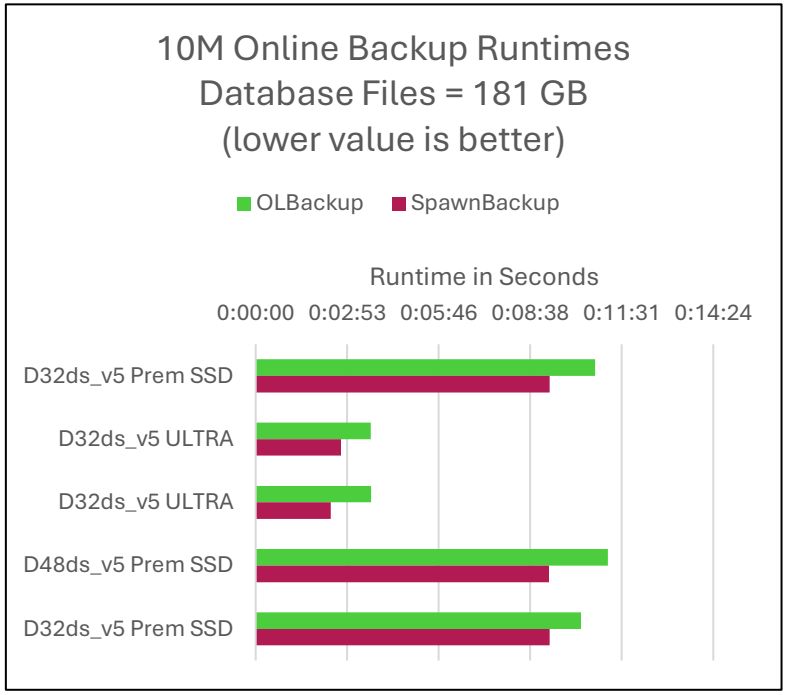


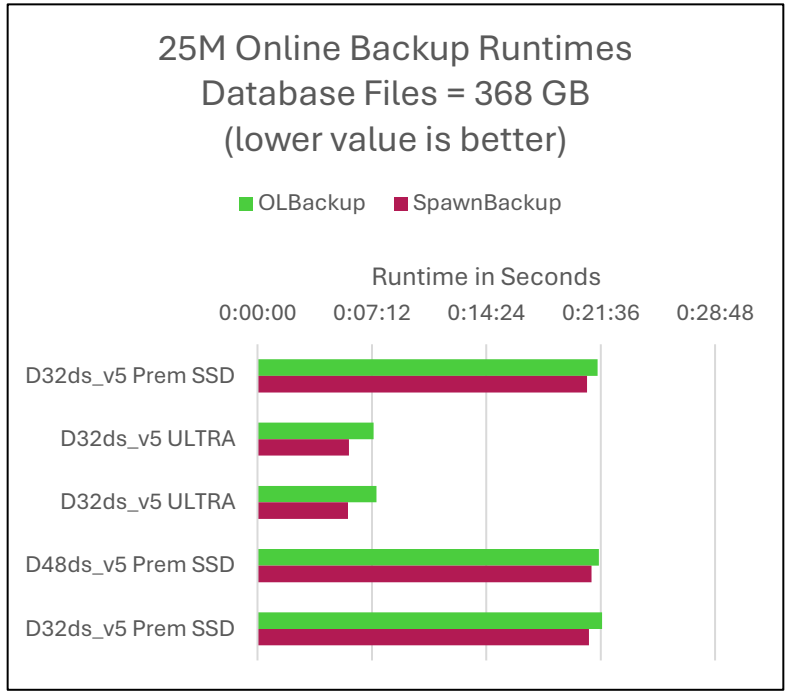


Online Backup Summary

- sca_olbackup.sh – backup all regions of a GT.M database
- sca_pa_spawn_olbackup.sh – multi-threaded online backup
- GT.M mupip utility backup

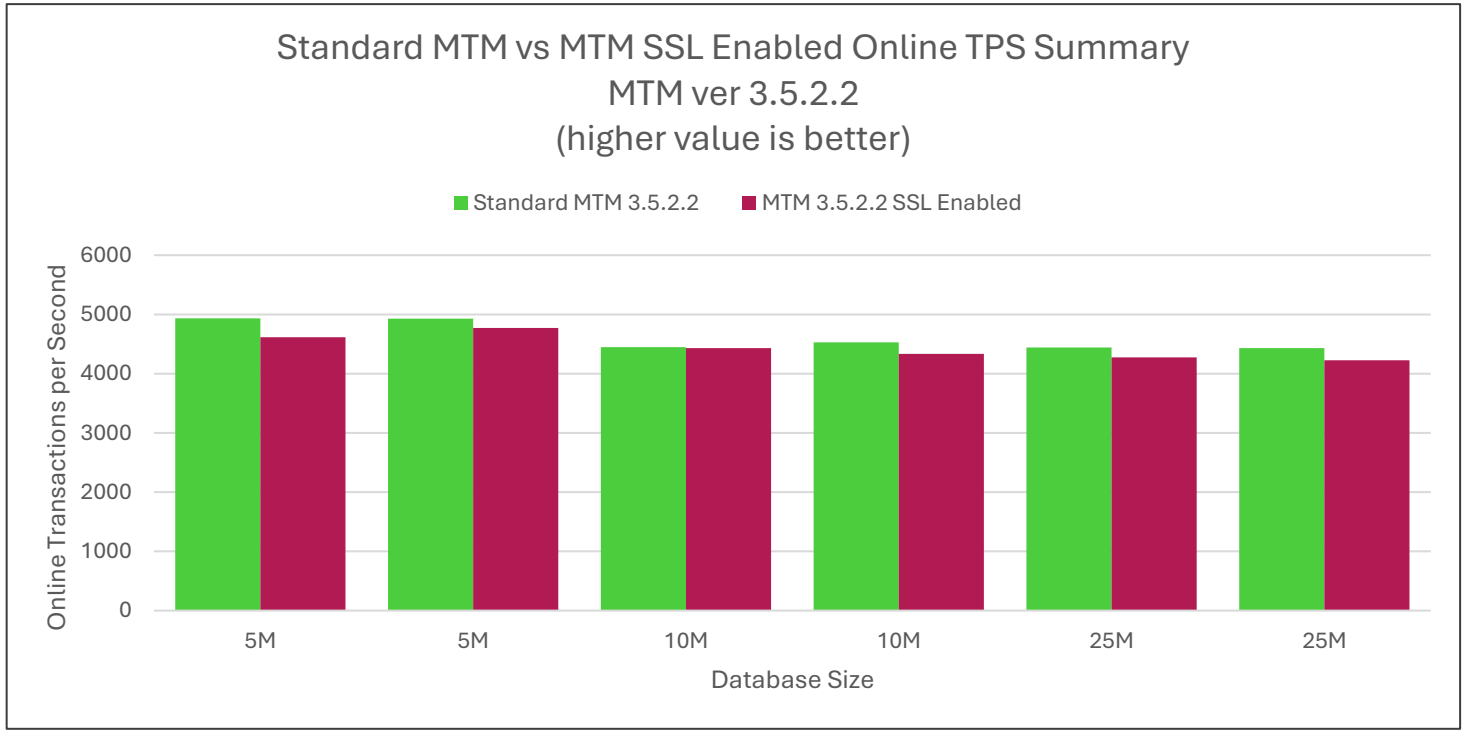






MTM-SSL Enabled Findings

- Linux64 MTM version 3.5.2.2
- TLSv1.2 Secure Communication
- On Average, MTM-SSL showed ~4% decrease in overall online performance



Next Steps / Future Performance Testing

- Finalize Detailed Benchmark Report
- Azure Cloud Profile Replication Testing
  - Replication timings and backlog levels
  - Local secondary instances setup
  - DR secondary instances setup
- Profile core read-only instance testing
  - Document configuration and performance setups
  - Evaluate functional and performance characteristics
- Azure Console Monitoring Tools Review and Testing
- Testing Higher End AMD Processor VMs
- Evaluate/Optimize GT.M Encryption Plug-in Performance

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